



Optimization of Film Flexibility and Structural Integrity in Marine-Derived Biopolymers for Single-Use Cup Fabrication

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ABSTRACT

The proliferation in single-use plastic consumption has exacerbated severe environmental impacts, contributing to microplastic pollution, greenhouse gas emissions, and eutrophication in marine ecosystems. To address this, marine-derived biopolymer alternatives, such as kelp and seaweed, were explored as replacements for conventional land-based bioplastics. This study aims to develop and optimize a biodegradable single-use cup, experimenting with kelp and seaweed extracts. Mixtures of biopolymer extracts will be combined with gelatin and starch to enhance structural cohesion, and the compatibility with glycerol as a plasticizer will be explored to test improvements in flexibility. In this study, mixtures of kelp or seaweed (10% w/v) were combined with gelatin or starch (40% w/v) and tested for four dependent variables: hardening rate, brittleness, surface appearance, and structural integrity. The mixtures were also tested under various drying conditions with the addition of glycerol (2% v/v) as a plasticizer, and their effects on flexibility, cracking, structural integrity, and surface appearance were evaluated. The kelp (40% w/v) and gelatin (10% w/v) blend, when dried under controlled conditions in the absence of glycerol, produced the most structurally integral, durable, and smooth cup with minimal surface defects. These findings reveal the potential of marine-derived biopolymers as a scalable, eco-friendly, and economically accessible material for the synthesis of sustainable single-use cups. It offers a sustainable alternative to typical materials such as plastic and plant-based polymers.

KEYWORDS

Biodegradable polymers, Film flexibility, Gelatin reinforcement, Kelp-based bioplastics, Single-use cup fabrication

INTRODUCTION

Despite predicted improvements in recycling rates, amounts of plastic waste will nearly double by 2050, increasing from 464 Mt in 2020 to approximately 735.4 Mt per year by 2050 ("Global Projections of Plastic Use, End-of-Life Fate and Potential Changes in Consumption, Reduction, Recycling and Replacement with Bioplastics to 2050," 2024). The global Single-use Plastic (SUP) Market is expected to increase from a valuation of \$58.4 billion to \$83.8 billion from 2023 to 2032, recording a compound annual growth rate of 4.1% (*Global Single Use Plastic Market Size, Trends, Share 2032*, n.d.). Most SUPs are incinerated or end up in landfills, accumulating in oceans, rivers and soils ("Single-Use Plastics: Production, Usage, Disposal, and Adverse Impacts," 2021). Furthermore, SUPs often degrade slowly into microscopic fragments called secondary microplastics (MP) as a result of UV rays or heat, making plastic pollution more difficult to track ("Single-Use Plastics: Production, Usage, Disposal, and Adverse Impacts," 2021). SUP waste can enter storm drains or sewage systems, causing ingestion of plastics for

humans and animals in the environment ("Single-Use Plastics: Production, Usage, Disposal, and Adverse Impacts," 2021). Incineration of plastics emits greenhouse gases and toxic pollution that contribute to climate change. According to an OECD estimate, plastics contribute approximately 3.3% of global CO₂ emissions, resulting in 1.8 billion tonnes of lifecycle emissions of plastics (Ritchie, 2023).

Profitable due to their convenience and low manufacturing costs, SUPs are widely used for plastic cups, especially in coffee shops and the fast-food industry. Approximately 5.8 billion SUP cups were consumed just in coffee shops and fast-food restaurants in South Korea in 2022 (Lee et al., 2025). One of the most explored solutions to SUP cups are bioplastics such as polylactic acid (PLA) (Rezvani et al., 2021). PLAs, produced from renewable plant-based sources such as corn, cassava, or sugarcane, produce 68% fewer greenhouse gas emissions than conventional plastics without any toxins (*What Is PLA? (Everything You Need To Know)*, n.d.). However, the decomposition of PLA still requires special composting in facilities that require heat

of up to 140°C for 10 days, and the natural breakdown of PLA would take approximately 100 years (Rezvani et al., 2021). Other potential solutions include disposable paper cups; however, many of these cups contain hydrophobic films made of polyethylene plastics that can deteriorate when exposed to hot water (Ranjan et al., 2021). Studies have shown that microplastic particles, along with ions such as fluoride and chloride, and heavy metals, can leach into beverages when exposed to temperatures of 85-90°C, making them a potentially harmful alternative to conventional SUPs (Ranjan et al., 2021).

While a currently far more unimplemented solution, macroalgae (e.g., seaweed/kelp) has emerged as a potential replacement for conventional SUPs. They have high mechanical strength suited for a wide range of processes in varying sizes and structures, and the synthesis of biofilm from seaweed polysaccharides can provide protection against bacteria, UV rays, and oxidation (*Website*, n.d.). Seaweed-based plastic has certain environmental and economic advantages over commonly used PLAs and paper alternatives. Its cultivation is fast-paced and is not dependent on arable land, fertilizers, or freshwater, leading to net reductions in eutrophication in freshwater and marine ecosystems—a major issue associated with fossil-based plastics (“Prospective LCA of Brown Seaweed-Based Bioplastic: Upscaling from Pilot to Industrial Scale,” 2024). An industrial-scale life-cycle assessment upscaling seaweed production and simulating fabrication of seaweed-based plastics showed that 1 kg of brown seaweed-based bioplastic had a net +1.37kg CO₂-eq climate impact (including biogenic carbons), far less than the impact of low-density polyethylene of +3.6kg CO₂-eq (“Prospective LCA of Brown Seaweed-Based Bioplastic: Upscaling from Pilot to Industrial Scale,” 2024). This indicates that alginate bioplastics can potentially serve as a far more environmentally sustainable material for SUP cups than traditional methods. Beyond environmental gains, a techno-economic assessment found that calcium-alginate seaweed films can be produced at \$4.56/kg, and it has an ultra-low oxygen-barrier performance, requiring far less plastic material for packaging than PLA and PET (Mohammed et al., 2023).

In this research, we develop a biodegradable single-use cup using seaweed and kelp extracts as primary natural macroalgae polymers. We hypothesize that seaweed and kelp polysaccharides may provide structural strength properties to form the cup, and gelatin is hypothesized to enhance the flexibility, durability, and structural cohesion of the cup. The mixture may form a stable biopolymer suitable for single-use cup fabrication, showing enhanced water resistance and mechanical strength when compared to individual extracts. This study hypothesizes that marine-derived macroalgae biopolymers combined with natural additives may result in a fully biodegradable and functional single-use cup providing an environmentally sustainable and cost-efficient potential alternative to conventional SUPs.

METHODS

Materials

Dried dashi kelp powder, dried seaweed powder, gelatin, and starch granules were used as the primary raw materials for the preparation of biodegradable films. Distilled water was used as the solvent. All chemicals were of food-grade or analytical grade.

Preparation of Stock Solutions

Two types of stock extracts were prepared. For the dashi kelp extract, 5 g of kelp powder was mixed with 50 g of distilled water and stirred until fully dissolved. Similarly, for the seaweed extract, 5 g of seaweed powder was dissolved in 50 mL of distilled water. For larger-scale preparations, 20 g of kelp or seaweed powder was added.

Sample Preparation

To evaluate the effects of different additives on film formation, six experimental conditions were established using both dashi kelp and seaweed extracts. Each condition was prepared in a final volume of 10 mL. The first group consisted of dashi kelp extract alone, dashi kelp extract supplemented with 1 g of gelatin, and dashi kelp extract supplemented with 1 g of starch granules. The second group consisted of seaweed extract alone, seaweed extract supplemented with 1 g of gelatin, and seaweed extract supplemented with 1 g of starch granules. These variations allowed for the comparison of additive-free extracts with those reinforced by protein (gelatin) or polysaccharide (starch) components.

Mixing and Casting

Each mixture was gently heated to approximately 60 °C while being stirred until the gelatin or starch had dissolved entirely and a homogeneous solution was obtained. The hot solutions were then poured into sterile Petri dishes or silicone molds.

Characterization

The resulting films were visually inspected for texture, thickness, uniformity, and transparency. Flexibility and mechanical strength were compared qualitatively among the six conditions. Water resistance was preliminarily evaluated by adding droplets of water onto the films and recording dissolution or swelling behavior.

RESULTS AND DISCUSSION

Film Formation and Initial Observations

Table 1. Film formation and initial observations of various combinations of materials after two days of drying.

Material Base	Hardening Rate	Brittleness /Cracks	Surface Appearance	Cohesion / Structural Integrity
Seaweed (10% w/v)	Medium	No visible cracks	Glossy, smooth, greenish brown	Soft, tears easily
Kelp (10% w/v)	Medium	No visible cracks	Smooth, shinier, dark brown	Soft, tears easily
Seaweed (10% w/v) + Potato starch (40% w/v)	Fast	Brittle, Large visible cracks & small tears throughout	Very dry, brown with white specks	Soft, malleable
Seaweed (10% w/v) + Gelatin (40% w/v)	Slow	No visible cracks	Small bubbles, shiny	Liquid, viscous
Kelp (10% w/v) + Potato starch (40% w/v)	Fast	Brittle, Small tears, small circular holes at center	Dry, smooth, holes present,	Soft, malleable
Kelp (10% w/v) + Gelatin (40% w/v)	Slow	No visible cracks	More white bubbles, shinier	Liquid, viscous

Table 1 summarizes qualitative differences between materials used for the cup after two days of drying. The trials used kelp and seaweed as base materials, tested with 40% concentrations (by mass) of both potato starch and gelatin. Water mixtures of kelp/seaweed that did not include potato starch or gelatin were used as controls.

Potato starch and gelatin revealed differences in hardening rates. Potato starch-based films hardened faster compared to gelatin-based films, but they also exhibited relative brittleness with cracks across the surface. Gelatin-based films took much longer to harden and showed no visible cracks. The control groups without gelatin or potato starch revealed a smooth surface, but were very delicate to the touch.

The two variables also had differences in surface and consistency, with gelatin-based films resulting in a shinier surface with bubbles, while potato starch-based films had a dry surface with many holes present on their surface.

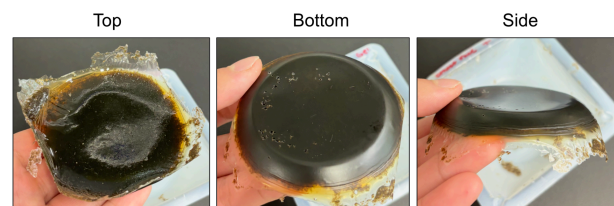


Figure 1. Image of optimized Kelp (10% w/v) + Gelatin (40% w/v) from top, bottom, and side views.

After 4 weeks of drying, the film with 10% kelp and 40% gelatin concentration exhibited the most cohesive solid structure. The top revealed a glass-like surface, while the bottom and sides resulted in matte-like surfaces.



Figure 2. Image of creating a cup using optimized Kelp (10% w/v) + Gelatin (40% w/v) in rapid drying conditions.

Rapid drying conditions failed to generate the cup shape with optimized variables (Kelp 10% w/v with Gelatin 40% w/v) determined from Figure 1. The mixture failed to form a uniform structure within the cup mold. The bubbles on the surface of the mold created undesirable dry foam. The resulting mixture was gelatinous and inflexible (tore easily).

Optimization of Film Flexibility With Controlled Drying

Table 2. Optimization of Film Flexibility with Controlled Drying

Condition	Variables Tested	Effect on Flexibility	Effect on Cracking	Notes on Surface/ Structure
Rapid Drying ($\geq 70^{\circ}\text{C}$), (refer Figure 2.)	High oven temperature, fast drying, no glycerol	Limited flexibility	High crack formation	Internal stress, uneven films
No Plasticizer + Controlled Drying	Baseline (no glycerol)	Low flexibility	Moderate reduction in cracks	Rigid films, moderate holes on surface
Glycerol (2%) + Controlled Drying	Moderate addition	Increased flexibility	Moderate reduction in cracks	Softer films, surface appeared greasy and

				leaky, bubbles forming holes on surface
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To improve the mixture’s flexibility and cohesion in the cup mold, controlled drying (dried at room temperature) was used along with the addition of Glycerol (2%) as a plasticizer. Table 2 shows that the plasticizer improved overall flexibility and reduced surface cracks on the mold. The glycerol significantly increased the flexibility of the film, but created a greasy and leaky surface. The film’s surface had many holes from bubbles in the mixture. Overall, the mix with no glycerol provided greater rigidity and formed fewer holes on the film’s surface.

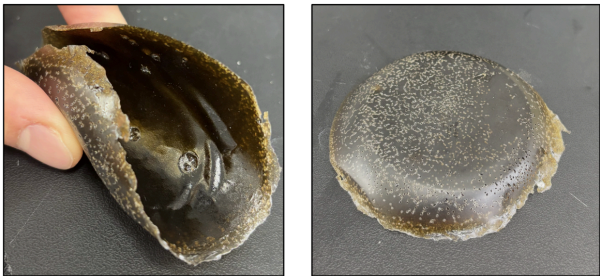


Figure 3. Image of Glycerol (2%) + Controlled Drying.

The addition of glycerol significantly increased flexibility, but created a greasy surface with numerous trapped bubbles and holes.

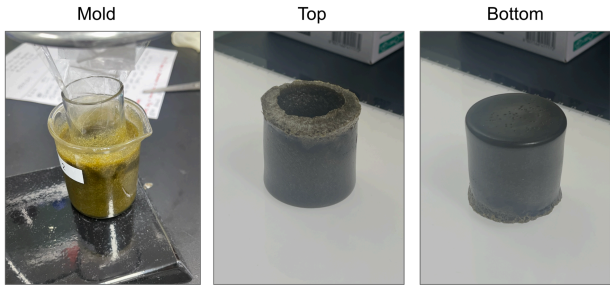


Figure 4. Image of cup shape mold (left), the image of cup in top view (middle), and bottom view (right).

Using a mixture of Kelp (10% w/v) and Gelatin (40% w/v) with controlled drying and no additional plasticizers, the final cup was produced. The product resulted in strong cohesion and minimal surface defects. The top of the product had a rigid texture, which required further surface polishing with abrasives.

DISCUSSION

Study Impact

The results of this study demonstrate promising potential for marine-derived biopolymers as a sustainable and eco-friendly alternative to SUPs. Unlike predominant

land-based bioplastics such as PLA, the resulting kelp-gelatin biopolymer is easily biodegradable, eliminating the need for specialized composting facilities that require specific thermal conditions for decomposition ((Rezvani et al., 2021). The cultivation of kelp is environmentally restorative as well, as it mitigates eutrophication, an issue exacerbated by cultivating plant-based polymers through terrestrial agriculture (“Prospective LCA of Brown Seaweed-Based Bioplastic: Upscaling from Pilot to Industrial Scale,” 2024). Large-scale production of kelp can help mitigate the impacts of eutrophication and provide suitable biopolymer alternatives simultaneously. Furthermore, kelp is a fast-growing species, making it economically viable and scalable for the production of biopolymers when paired with efforts to reverse nutrient runoff (Mohammed et al., 2023). As such, we conclude that kelp-based biopolymers can provide a viable pathway for fabricating single-use cups that align with environmental conservation.

Limitations and Future Plan

While the study’s findings show a promising solution, several limitations must be considered for future research. Although the final prototype demonstrated moderate cohesion, it still exhibited small surface pores and irregularities during drying, potentially due to air entrapment. Further surface optimization techniques, such as vacuum degassing, could be explored to create even drying by removing trapped air (“Comparative Analysis of Degassing Methods for Preparation of Pinhole-Free Membranes,” 2025). The cup’s mechanical durability was also not quantifiably tested, so further testing under high heat and pressure can be conducted. Additionally, as the film is derived from dried marine algae and gelatin, it can release more odors than other preexisting single-use cup polymers (“Bioplastic Feedstock Production from Microalgae with Fuel Co-Products: A Techno-Economic and Life Cycle Impact Assessment,” 2020). Overall, smaller increments of gelatin-kelp ratios, or the addition of other biodegradable agents, could be tested to improve the optimization of odor reduction, enhance surface features, and increase cohesion.

CONCLUSIONS

The study explored marine biopolymers, such as kelp and seaweed, as alternative base materials for the synthesis of single-use cups. The findings show that a water mixture of kelp (10% w/v) and gelatin (40% w/v) can form cohesive, flexible, and biodegradable films suitable for single-use cup fabrication that are sustainable and economically viable alternatives to conventional plastics and plant-based biopolymers such as PLA. Controlled drying conditions produced a stable structure, while excessive heat or the addition of plasticizer reduces surface quality. Although further optimization is necessary to enhance surface uniformity, odor, and mechanical durability, the results indicate a promising potential for the future development of marine-sourced biodegradable materials for eco-friendly single-use packaging.

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